

G. Yash And Trees

time limit per test: 4 seconds
memory limit per test: 512 megabytes

Yash loves playing with trees and gets especially excited when they have something to do with prime numbers. On his 20th birthday he was granted with a rooted tree of n nodes to answer queries on. Hearing of prime numbers on trees, Yash gets too intoxicated with excitement and asks you to help out and answer queries on trees for him. Tree is rooted at node 1. Each node i has some value a_i associated with it. Also, integer m is given.

There are queries of two types:

- for given node v and integer value x , increase all a_i in the subtree of node v by value x
- for given node v , find the number of prime numbers p less than m , for which there exists a node u in the subtree of v and a non-negative integer value k , such that $a_u = p + m \cdot k$.

Input

The first of the input contains two integers n and m ($1 \leq n \leq 100\,000$, $1 \leq m \leq 1000$) — the number of nodes in the tree and value m from the problem statement, respectively.

The second line consists of n integers a_i ($0 \leq a_i \leq 10^9$) — initial values of the nodes.

Then follow $n - 1$ lines that describe the tree. Each of them contains two integers u_i and v_i ($1 \leq u_i, v_i \leq n$) — indices of nodes connected by the i -th edge.

Next line contains a single integer q ($1 \leq q \leq 100\,000$) — the number of queries to proceed.

Each of the last q lines is either $1 \ v \ x$ or $2 \ v$ ($1 \leq v \leq n$, $0 \leq x \leq 10^9$), giving the query of the first or the second type, respectively. It's guaranteed that there will be at least one query of the second type.

Output

For each of the queries of the second type print the number of suitable prime numbers.

Examples

input	Copy
8 20 3 7 9 8 4 11 7 3 1 2 1 3 3 4 4 5 4 6 4 7 5 8 4 2 1 1 1 1 2 5 2 4	
output	Copy
3 1 1	

input	Copy
5 10 8 7 5 1 0 1 2 2 3 1 5 2 4 3 1 1 0 1 1 2 2 2	
output	Copy
2	

Manthan, Codefest 16

Finished

Practice

Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

Submit?

Language: GNU G++23 14.2 (64 bit, ms)

Choose file: Choose File No file chosen

Submit

Submission	Time	Verdict
362098348	Feb/09/2026 11:47	Accepted
362097705	Feb/09/2026 11:41	Wrong answer on test 54
362097422	Feb/09/2026 11:38	Wrong answer on test 54
362097305	Feb/09/2026 11:37	Wrong answer on test 3
362097025	Feb/09/2026 11:34	Wrong answer on test 3

Problem tags

bitmasks data structures
dfs and similar math number theory
*2800

No tag edit access

Contest materials

Announcement (en)

Tutorial (en)